

Comprehensive IPv6 Architectural Guide & Journal Notes

Header Structures, Flow Control, Extension Headers, Addressing Paradigms, and Transition Mechanisms

Computer Networking Study Notes

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Abstract

This document serves as an exhaustive, publication-grade academic note detailing the Internet Protocol Version 6 (IPv6) architecture. It addresses core protocol concepts, including the fixed 40-byte base header, flow label-driven packet routing, the linked-list structure of extension headers, bit-level layout of unicast/multicast/anycast addresses, and IPv4-to-IPv6 transition mechanisms (Dual Stack, Tunneling, and NAT-PT/NAT64).

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1 IPv6 Base Header Architecture & IPv4 Equivalences

1.1 Key Protocol Modifications & Core Parameters

- **Fixed Header Size:** The standard IPv6 base header is strictly fixed at 40 **bytes** (320 **bits**), unlike the IPv4 header which ranges from 20 to 60 bytes due to optional fields. *Correction Note:* The raw lecture transcript mentioned "40 bits"; in standard networking terminology, the base IPv6 header is 40 **bytes**.
- **Address Length:** IPv6 addresses are 128 **bits** (16 bytes) in length, expanding the address space from IPv4's $2^{32} \approx 4.3 \times 10^9$ to $2^{128} \approx 3.4 \times 10^{38}$ unique addresses.
- **No Intermediate Fragmentation:** Unlike IPv4, intermediate routers in IPv6 **do not perform packet fragmentation**. Fragmentation is handled exclusively by the sending host using Path MTU Discovery (PMTUD) and an optional Fragment Extension Header.
- **Field Nomenclature Equivalences:**

Table 1: IPv4 vs. IPv6 Header Field Nomenclature

IPv4 Field	IPv6 Field	Functional Purpose
Time to Live (TTL)	Hop Limit	Decrement by 1 at each router; packet dropped if 0.
Type of Service (TOS)	Traffic Class	Classifies packet priority and QoS / DiffServ.
Total Length	Payload Length	Size of data following the 40-byte base header (max $2^{16} - 1$ B).
Protocol	Next Header	Specifies protocol inside payload or next extension header.
Options & Padding	Extension Headers	Additional control info appended outside base header.

1.2 The Flow Label Mechanism: 5-Tuple Fast-Path Routing

The Concept & Analogy: Imagine a highway toll booth where the first car in a fleet provides driver credentials, destination passes, and payment details. The operator inspects everything, calculates the best route, and hands the driver a colored pass (a **Flow Label**). When the next 4 cars in the exact same fleet arrive, the toll operator does not re-check their credentials or re-calculate routes; seeing the matching flow pass, the operator immediately waves them through onto the pre-assigned fast lane.

1.2.1 Technical Workflow

A network **Flow** is identified by a unique **5-tuple**:

$$\text{Flow 5-Tuple} = \{\text{Source IP, Destination IP, Source Port, Destination Port, Transport Protocol}\} \quad (1)$$

1. **Packet 1 Arrival:** When the initial packet of a communication flow reaches a router, the router inspects all 5 parameters. It performs routing table lookup, Quality of Service (QoS) evaluation, and path selection calculations.
2. **Flow Identification:** The router assigns a unique 20-bit **Flow Label** (generated by the source or ingress router) associated with this calculated path in its internal switching cache.
3. **Subsequent Packets (Packets 2-N):** When subsequent packets belonging to the same 5-tuple arrive, the router inspects the 20-bit **Flow Label** in the fixed IPv6 header.

4. **Fast-Path Forwarding:** The router bypasses full routing table lookup, deep packet inspection, and path computation, immediately forwarding the packet along the cached path. This dramatically reduces packet processing overhead and latency.

2 Extension Headers & Linked-List Architecture

2.1 Next Header Protocol Identification

The `Next Header` field in IPv6 serves a dual purpose: identifying upper-layer transport protocols or pointing to an Extension Header (EH).

- **TCP Protocol Identifier: 6**
- **UDP Protocol Identifier: 17**

Correction Note: The transcript queried whether TCP is 6 and UDP is 17 or vice versa. In the IANA protocol number registry, ****TCP is 6**** and ****UDP is 17****.

2.2 Linked-List Extension Header Chain

Instead of expanding the base header with optional fields (like IPv4), IPv6 inserts optional ****Extension Headers**** between the 40-byte base header and the upper-layer payload (UL). Each extension header contains its own `Next Header` field, forming a unidirectional **linked list**.

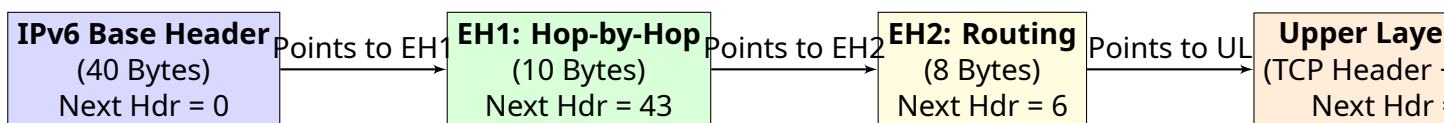


Figure 1: IPv6 Extension Header Linked List Structure.

2.3 Key Types of Extension Headers

1. **Hop-by-Hop Options Header (Value: 0):** Carried information that must be examined by **every router** along the path (e.g., Jumbo Payload option for packets > 65,535 bytes).
2. **Routing Header (Value: 43):** Specifies one or more intermediate nodes that the packet must visit on its way to the destination (source routing).
3. **Fragment Header (Value: 44):** Used when a source host needs to send a packet larger than the path MTU. Contains a unique Fragment Identification payload so the destination host can reassemble original fragments.
4. **Destination Options Header (Value: 60):** Carries information intended strictly for the final destination node (e.g., mobile IPv6 care-of address binding).
5. **Authentication Header (AH, Value: 51):** Part of IPsec; provides data integrity, source authentication, and anti-replay protection without encryption.
6. **Encapsulating Security Payload (ESP, Value: 50):** Part of IPsec; provides data confidentiality (encryption), payload integrity, and authentication.

3 IPv6 Address Architecture & Classification

IPv6 addresses are categorized into three main operational paradigms:

- **Unicast (1-to-1):** Identifies a single network interface. A packet sent to a unicast address is delivered to exactly one interface.
- **Multicast (1-to-Many):** Identifies a group of network interfaces. A packet sent to a multicast address is delivered to **all interfaces** registered to that multicast group. **Note: IPv6 has NO broadcast addressing;** multicast replaces broadcast.
- **Anycast (1-to-Nearest):** Assigned to multiple interfaces (typically belonging to different nodes). A packet sent to an anycast address is routed to the **single nearest interface** according to the routing protocol metric (commonly used in Content Delivery Networks / CDNs).

3.1 Master Unicast Address Prefix Table

Below is the classification table directly corresponding to standard IPv6 address allocations (as highlighted in lecture slides):

Table 2: Master Classification of IPv6 Unicast Address Space

Block Prefix (Binary)	CIDR Prefix	Block Assignment	Fraction of Space
0000 0000	0000::/8	Special Addresses (Unspecified, Loopback)	1/256
001	2000::/3	Global Unicast Addresses (Public IPs)	1/8
1111 110	FC00::/7	Unique Local Unicast (Private IPs)	1/128
1111 1110 10	FE80::/10	Link-Local Addresses	1/1024
1111 1111	FF00::/8	Multicast Addresses	1/256

3.2 Scope Hierarchy of Unicast Addresses

The scope of an IPv6 address determines its routing reachability across network boundaries:

Link-Local Scope (FE80::/10) $\subset$ **Organization / Site Scope** (FC00::/7) $\subset$ **Global Scope** (2000::/3)

4 Detailed Bit-Level Breakdown of Unicast Addresses

4.1 Special Addresses

- **Unspecified Address (::/128):** 128 bits of zero (0000...0000). Used by hosts during initial boot before obtaining an IP address (e.g., in DHCPv6 solicitations).
- **Loopback Address (::1/128):** 127 bits of zero followed by 1 bit of one. Used by a host to send packets to itself for local testing.
- **IPv4-Mapped IPv6 Address (::ffff:0:0/96):** Formatted as 80 zero bits, followed by 16 one bits (FFFF), followed by 32 bits of the standard IPv4 address. Used by dual-stack nodes to represent IPv4 endpoints in an IPv6 format.

4.2 Global Unicast Address (GUA) & EUI-64 Generation

Global Unicast Addresses are equivalent to public IPv4 addresses and are globally routable across the public Internet.

Table 3: Global Unicast Address (2000 :: /3) Bit Breakdown

Global Routing Prefix	Subnet ID	Interface ID
48 Bits	16 Bits	64 Bits
<i>Assigned by ISP/RIR</i>	<i>Internal Subnet Identifier</i>	<i>Host Identifier (EUI-64 or Random)</i>

4.2.1 The EUI-64 Method for Interface ID Generation

To auto-configure a 64-bit Interface ID using a hardware MAC address (48 bits):

1. Take the 48-bit MAC address (e.g., 00:1A:2B:3C:4D:5E).
2. Split the MAC address in half: 00-1A-2B and 3C-4D-5E.
3. Insert the fixed 16-bit hex value FF-FE in the middle: 00-1A-2B-FF-FE-3C-4D-5E.
4. Invert the **7th bit** (Universal/Local bit, U/L bit) of the first byte:

$$\text{First byte: } 00_{16} = 00000000_2 \xrightarrow{\text{Invert 7th Bit}} 00000010_2 = 02_{16} \quad (2)$$

5. Resulting EUI-64 Interface ID: 021A:2BFF:FE3C:4D5E.

4.3 Unique Local Unicast Address (ULA: FC00 :: /7)

Unique Local Addresses are equivalent to IPv4 private addresses (e.g., 10.0.0.0/8, 192.168.0.0/16). They are routable across private subnets/sites, but **never forwarded on the public Internet**.

Table 4: Unique Local Address Bit Structure (FC00 :: /7 / FD00 :: /8)

Prefix (7 bits)	L Bit (1 bit)	Global ID (40 bits)	Subnet ID (16 bits)	Interface ID (64 bits)
1111 110	1 (Locally Assigned)	Pseudo-Random Number	Custom Subnet	MAC / EUI-64 Identifier

- **Example ULA Prefix:** FD12:3456:789A:0001 :: /64
- **Key Feature:** The 40-bit pseudo-random global ID ensures that if two private organization networks merge, IP collision probabilities are extremely low.

4.4 Link-Local Address (LLA: FE80 :: /10)

Link-Local Addresses are auto-configured on every IPv6 interface. They are used exclusively for single-link communication (e.g., router discovery, neighbor discovery, DHCPv6 communications).

Difference Between ULA and LLA:

- **Link-Local Address (FE80 :: /10):** Routers **WILL NOT** forward packets with LLA beyond the local link (switch layer).
- **Unique Local Address (FC00 :: /7):** Routers **WILL** forward ULA packets between in-

Table 5: Link-Local Address Bit Structure (FE80::/10)

Prefix (10 bits)	Zero Bits (54 bits)	Interface ID (64 bits)
1111 1110 10	0000...0000	Generated via EUI-64 or Random

ternal corporate site subnets, but edge routers drop them before reaching the public Internet.

5 Multicast & Anycast Addressing

5.1 Multicast Address Structure (FF00::/8)

IPv6 multicast addresses start with the prefix byte FF (1111 1111₂).

Table 6: Multicast Address Layout

Prefix (8 bits)	Flags (4 bits)	Scope (4 bits)	Group ID (112 bits)
1111 1111	0 0 T P	Scope Identifier	Unique Multicast Group Identifier

5.1.1 Flag Bits breakdown

- **T (Transient Bit):** 0 = Permanent / Well-known multicast address (assigned by IANA); 1 = Dynamically assigned / Transient.

5.1.2 Scope Bits Table

Table 7: Multicast Scope Field Values

Scope Value (Hex)	Scope Name	Reachability
1	Interface-Local	Does not leave local interface (loopback).
2	Link-Local	Confined to the local physical network segment.
4	Admin-Local	Confined to administrative boundary set by router.
5	Site-Local	Confined to internal site/campus network.
8	Organization-Local	Covers all networks owned by an organization.
E	Global	Unrestricted globally across Internet.

5.1.3 Practical Multicast Examples for Students

- **FF02::1 (All-Nodes Link-Local):** Reaches every IPv6 host on the local physical link. (Replaces IPv4 subnet broadcast address 255.255.255.255).
- **FF02::2 (All-Routers Link-Local):** Reaches all IPv6 routers connected to the local link.

5.2 Anycast Addressing Mechanism

An Anycast address is syntactically indistinguishable from a standard Unicast address. However, it is assigned to **multiple interfaces** across different topological locations.

- **CDN Application:** A global DNS server or Content Delivery Network (e.g., Cloudflare 1.1.1.1 or IPv6 equivalent) assigns the same Anycast IP to servers in Tokyo, London, and New York.
- BGP routing automatically directs client traffic to the **topologically nearest server**, minimizing latency.

6 IPv4 to IPv6 Transition Mechanisms

Because replacing global IPv4 infrastructure overnight is impossible, three primary coexistence techniques are deployed:

6.1 1. Dual Stack Strategy

Nodes run both IPv4 and IPv6 protocol stacks simultaneously. The operating system uses DNS queries (A for IPv4, AAAA for IPv6) to determine whether to communicate over IPv4 or IPv6.

6.2 2. IPv6 Tunneling (6in4 / Teredo / GRE)

When two IPv6 networks are separated by an intermediate IPv4-only network:

1. The entry dual-stack router takes the IPv6 packet (40-byte IPv6 header + payload).
2. It **encapsulates** the IPv6 packet inside a standard 20-byte IPv4 header (IP Protocol type 41).
3. The packet travels across the IPv4 internet as standard IPv4 traffic.
4. The receiving dual-stack router **decapsulates** the packet, removes the IPv4 header, and routes the original IPv6 packet into the destination IPv6 network.

6.3 3. Network Address Translation – Protocol Translation (NAT64 / NAT-PT)

Used when an IPv6-only host needs to communicate with an IPv4-only server. The NAT64 gateway translates IP header fields (mapping 128-bit IPv6 addresses to 32-bit IPv4 addresses) and translates transport protocol headers in real time.